



Overview of Sport-Specific Injuries

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The intricacies of many sports result in patterns of injury that are different compared with those of the general athletic population. It is important that health care providers caring for professional and nonprofessional athletes are familiar with their specific needs to afford accurate recognition and swift treatment. Sports injuries have been well documented in high school and collegiate settings.¹ Overall, injury rates appear to be higher in competition than in practice. While non-time-loss injuries account for more than half of all injuries, the remainder can be devastating to the athlete and difficult to treat. Lower extremity injuries account for about half of all injuries at both the high school and collegiate levels.¹ Ankle sprains are the most common. Though anterior cruciate ligament (ACL) injuries are infrequent (3% of all lower extremity injuries), growing interest exists due to their cause for significant time loss from sport. Upper extremity injuries comprise about one quarter of all injuries at both levels, with shoulder injuries most common. Head and neck injuries are low, about 10% of all sporting-related injuries. Concussions represent 4% to 6% of all injuries, with men's wrestling, hockey, and football; and women's soccer, basketball, and lacrosse causing the vast majority.¹ Of the nearly 12,000 spinal cord injuries per year in the United States, 10% are related to athletic activity.¹ From 1982 to 2013, 2101 catastrophic spinal cord injuries occurred from athletic activity and more than 80% of these injuries occur in high school athletes as compared with collegiate athletes.¹

Medical issues such as sudden cardiac arrest and death further plague athletes, with a reported incidence of 0.5 to 2.5/100,000 athlete-years.² Among collegiate athletes, sudden cardiac arrest was reported in 1:43,770 athletes per year, and 1:3000 athletes per year in African American Division 1 basketball players.² Though not commonplace in the United States, across Europe and elsewhere, electrocardiograms (ECGs) are gaining favor in preparticipation athletic evaluations, as it is estimated that over two-thirds of common causes of sudden cardiac death can be detected by ECG.²

The continued prevalence of injury and illness in the athletic population elucidates the need for team physicians to be knowledgeable of the issues, to provide the best care to athletes at all levels. This chapter will highlight the common injuries facing many sports in specific, and their potential causes based on each sport.

BASEBALL

Baseball is nicknamed "America's Pastime," which supports its status as the second-most commonly played team sport in the United States, with about 19 million people involved in playing every year.^{3,4} It is a ball-and-bat sport played by two teams with nine players on each team. The goal is to score "runs" by hitting a thrown ball and advancing around four bases. The game is broken up into nine sections ("innings"), during which each team has an opportunity to play offense ("at bat") and defense (in the field). Players are categorized by the position they play in the field. The pitcher is the person who throws the ball for the batter from the opposing team to hit. The catcher is the person who receives the ball from the pitcher. Basemen can be first, second, or third basemen, and they predominately position themselves around these bases. The "shortstop" plays between the second and third basemen. Outfielders are deep in the field and play the farthest away from the pitcher. The batter is on the opposite team, and becomes a runner upon successfully hitting the ball and advancing to one of the three bases.

Most baseball injuries are by nonhuman contact (45%) or no contact at all (42%).⁵ Contact injuries consist of collisions with high-velocity balls, bats, bases, outfield walls, and the ground. In addition, running, throwing, and pitching can cause acute and overuse injuries.

Pitchers tend to sustain more injuries than other players, with predictable predilection for the upper extremity of course due to their role.⁶ The pitching motion is broken up into six phases: wind-up, stride, arm cocking, arm acceleration, arm deceleration, and follow-through.⁷ Maximal forces occur in the arm acceleration phase.⁸ Studies have demonstrated forces of up to 500 N at the lateral radiocapitellar joint, velocities as high as 7000 degrees per second during the late arm cocking and arm acceleration phases, and an estimated external rotation torque as high as 67 Nm.⁹

Shoulder Injuries

Injuries specific to the thrower's shoulder most commonly involve the labrum and undersurface of the rotator cuff.⁹ These injuries can be brought on by tissue changes in both the anterior and posterior glenohumeral capsule, that alter shoulder kinematics.⁹ Maximal stress to the rotator cuff is exerted during the

Abstract

The intricacies of many sports result in patterns of injury that are different compared with those of the general athletic population. It is important that health care providers caring for professional and nonprofessional athletes are familiar with their specific needs to afford accurate recognition and swift treatment. Sports injuries have been well documented. Overall, injury rates appear to be higher in competition than in practice. While non-time-loss injuries account for more than half of all injuries, the remainder can be devastating to the athlete and difficult to treat. Lower extremity injuries account for about half of all injuries at both the high school and collegiate levels. Ankle sprains are the most common. Though anterior cruciate ligament (ACL) injuries are infrequent, growing interest exists due to their cause for significant time loss from sport. Upper extremity injuries comprise about one quarter of all injuries at both levels, with shoulder injuries most common. Head and neck injuries are low, about 10% of all sporting-related injuries. Concussions represent 4% to 6% of all injuries, with men's wrestling, hockey, and football; and women's soccer, basketball, and lacrosse causing the vast majority. Of the nearly 12,000 spinal cord injuries per year in the United States, 10% are related to athletic activity. From 1982 to 2013, 2101 catastrophic spinal cord injuries occurred from athletic activity and over 80% of these injuries occur in high school athletes as compared with collegiate athletes. In this chapter, we will discuss injuries specific to each sport, with a concise review of the sport, its characteristic activities, and risk factors for causing injury.

Keywords

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follow-through phase of throwing, as the cuff musculature functions to decelerate the arm.⁴ Since the superior labrum is the attachment site for the long head of the biceps tendon, repeated traction of the biceps tendon with overhead throwing can also lead to superior labrum anterior to posterior (SLAP) tears.⁴

Elbow Injuries

The throwing motion generates significant valgus stress on the elbow and rapid elbow extension. This combination places tension on the medial structures of the elbow, shear stress on the posterior structures, and compression forces on the lateral structures. The combination of this is described as valgus extension overload syndrome, and is responsible for the pathophysiology of most throwing elbow injuries.¹⁰ Medial epicondylitis is one sequela from the syndrome, in addition to ulnar collateral ligament (UCL) sprain or tear from overuse.¹⁰

Lower Extremity Injuries

The lower extremity is the most frequently injured body part among fielders. Upper leg muscle and tendon strains are the most common lower extremity injury, with hamstring strains being the most common.¹¹ Knee ligamentous injuries and ankle and foot sprains make up the other large numbers of lower extremity injuries.¹¹ Most of these injuries occur during sudden sprints, while sliding, or while manipulating one's body to catch a ball.

Young Athlete Injuries

Almost 5 million children and adolescents participate in baseball every year in the United States.³ Adolescent pitchers are at particular risk for overuse injuries from throwing.⁹ Studies have demonstrated a steady increase in surgical rates for pitching-related injuries in immature throwers.¹² In one study, pitching more than 100 innings per year demonstrated a significantly increased risk of injury in the adolescent population.¹² The elbow is most commonly affected, with a spectrum of pathology that includes an apophysitis or avulsion of the medial epicondyle and osteochondritis desiccans of the capitellum or radial head.¹⁰ In the shoulder, an epiphysiolysis of the proximal humerus may develop.¹⁰ Collectively, these conditions are known as "little leaguer's elbow" and "little leaguer's shoulder." Consensus recommendations include avoiding pitching with arm fatigue/pain, avoiding 80 pitches or more per game (2500 pitches or more per year), avoiding competitive pitching for longer than 8 months per year, and exercising caution and restraint in pitching showcases.¹³ In 2008, Little League Baseball guidelines disposed of inning restrictions for pitchers and moved to pitch counts, which were deemed more prognostic from their ongoing studies.¹⁴ Maximum allowable pitch counts per game are as follows: 50 pitches (age 7 to 8), 75 pitches (age 9 to 10), 85 pitches (age 11 to 12), 95 pitches (age 13 to 16), and 105 pitches (age 17 to 18).¹⁴ Depending on the amount of pitches thrown, mandatory rest is enforced as well. For example, if a pitcher aged 15 to 16 pitches 76 or more pitches in a single day, he must take four calendar days of rest before returning to pitching. The guidelines also suggest that youth pitchers take 4 months of rest per year, but this has not been mandated.¹⁴

Sliding Injuries

In the major league and collegiate levels, sliding injuries are low, accounting for less than 10% of all injuries.^{5,6,15} In a study of five seasons of Major League Baseball (MLB) play, an injury occurred in about 3 per 1000 slides.¹⁵ Average time loss was 12.3 days if no surgery was required, and 66.5 days if surgery was required, as was the case in 8.2% of injuries.¹⁵ Interestingly, players were almost four times more likely to be sliding into second base for their injury as compared with any other base.¹⁵ Feet-first slides most commonly caused ankle injury, and head-first slides most commonly caused injury to the hand, fingers, or thumb.¹⁵ At the collegiate level, the overall injury rate from sliding was 9.5 per 1000 slides, with feet-first slides resulting in double the amount of injuries as head-first slides.¹⁶ In stark contrast, sliding injuries may represent as many as 70% of all injuries sustained in recreational baseball and softball.¹⁷ Presently, low-impact or breakaway bases are available, and have been shown to significantly reduce the amount of injury caused by sliding.¹⁵

BASKETBALL

Basketball is a game played with five players on each team. The object is to score by shooting the ball into a hoop that is mounted 10 feet above the playing surface. The team with the ball is on offense, and the team without the ball is on defense. It is similar to soccer and hockey in that the possession of the ball can change teams rapidly. To advance the ball down the court, it must be dribbled by hand or passed from player to player. The five playing members usually have defined roles: point guard, shooting guard, forward, and center.

Basketball is an inherently vertical sport requiring 35 to 46 jumping and landing activities per game, constant acceleration/decelerations that are multidirectional in nature, and direction changes almost every 2 to 3 seconds.^{18,19} It's no surprise these contribute to its status as one of the leading sports causing injury, with an estimated half million physician visits per year.²⁰⁻²⁵ Injury rates are reportedly between 7 and 10 per 100,000 athlete exposures.²⁶ The most commonly reported injuries include ankle sprains, finger sprains/fractures, knee traumatic and overuse injuries, facial lacerations, dental injury, and concussion.²⁶

Ankle Sprains

Ankle sprains are the most common diagnosed injury in both male and female basketball players, accounting for 25% of all basketball injuries overall.^{27,28} Neuromuscular training and external ankle supports were found effective in significantly reducing ankle sprain incidence.²⁹ One study estimates that seven basketball players need to undergo neuromuscular training (consisting of a 9 week balance training program) for it to effectively prevent one ankle injury.³⁰ Ankle braces work by reducing the weight-bearing and non-weight-bearing inversion range of motion (ROM) at the ankle, increasing muscle activation and excitability, and decreasing joint velocity.³¹

Knee Injuries

ACL injuries are common in basketball. Females are two to four times more likely to sustain ACL rupture as compared with males,

with a 16% chance of occurring sometime during their career.³² Neuromuscular training has failed to show any tangible benefit in ACL rupture prevention in the basketball population in specific.³³⁻³⁵ ACL injury prevention programs are successful in other sports such as soccer, team handball, and volleyball.^{34,36}

Tendinitis

The patellar and Achilles tendons are vulnerable to overuse tendinopathies from the repetitive eccentric loading involved in jumping. Patellar tendinitis (“jumper’s knee”) is the most common overuse injury in basketball.³⁷ In junior basketball, its prevalence is reported as high as 10%.³⁷ Risk factors include high jumping, deep knee flexion during landing, and valgus strain during eccentric load phase of landing.³⁷

Finger/Hand

Injuries to the hand are very common in basketball.^{23,38} Most hand injuries (>90%) involve sprains and volar plate injuries of the proximal interphalangeal (PIP) and metacarpophalangeal joints.³⁹ Dislocations at the PIP joint are common from direct ball contact with axial load. Players are also at risk for avulsion of the extensor digitorum when the ball creates an axial load through the fingertip (“mallet finger”) and avulsion of the flexor digitorum profundus (FDP) when the finger is caught on an opponent’s jersey or on the rim during a slam dunk (“jersey finger”). Boutonnière deformity can result from rupture of the extensor tendon central slip; early recognition of this condition is important. Gamekeepers’ injuries to the thumb usually occur as a result of a fall to the floor or an extension load to the thumb while blocking an opponent.

Facial/Dental Injury

Facial injuries are more common in male basketball players as compared with female players.^{23,38} Dental injuries are composed of fractures, avulsions, and oral lacerations. Ocular trauma is common from contact with opponents’ fingers or elbows and can result in corneal abrasions, retinal detachments, hyphemas, lacerations, contusions, and fractures. Nasal fractures may occur, either from a blow by the opponent’s elbow or from head-to-head contact.

Concussion

Concussion is common in basketball, particularly during the physical contact of rebounding a ball under the basket. As in other sports a prior history of concussion is a predictor for repeat injury and should be evaluated during preseason examination. Every concussion should be managed on site and have medical clearance before returning to play.⁴⁰ Consensus guidelines exist to guide medical staff on recognizing and appropriately managing players with suspected concussion.⁴⁰

Sudden Death

Basketball is a high-intensity sport with moderate static and high dynamic cardiac demands. It is estimated that up to 35% of sudden deaths in sport occur in basketball. The majority of these deaths were classically thought to result from occult hypertrophic cardiomyopathy; however, more recent literature fails

to prove this at time of autopsy.⁴¹ Connective tissue disorders such as Marfan syndrome can also induce cardiac abnormalities. With a high preponderance of these disorders in tall people, and the predominantly tall basketball player population, it is important that adequate preparticipation screening with a cardiac-specific history is performed. Twelve-lead ECG and echocardiogram testing may prove useful in the correctly screened population and can potentially detect more than two-thirds of common causes of sudden cardiac death.^{2,39}

BOXING

Boxing is one of the most antique Olympic sports dating back to the ancient Greece. It is an individual sport where two pugilists engage in a gloved-fist fight in an enclosed ring. Amateur boxing is an Olympic sport as opposed to professional boxing, which is regulated by four international organizations. A match is usually composed of 3 to 12 rounds that are 3 minutes long, supervised by a referee who ensures the athletes’ safety as well as compliance with the rules of the match. The boxing match is won if an opponent is knocked down for a count of 10, if the referee disqualifies a boxer, if he judges that one of the opponent is incapable of protecting himself, or based on the decision of a panel of three judge after all rounds are completed.

Professional Versus Amateur

The majority of the epidemiological study in boxing is done in amateur boxing, where the majority of injuries involves the head and face.⁴² Amateur and professional boxing generally share similar rules, including that boxers can only be struck in the face or body; however, the differences in these two types of boxing explain variations in the type of injuries. One of the most obvious differences stems from the protective equipment. Amateur boxers wear protective headgear that shields the head and face from lacerations and orbital fractures.⁴³ In addition, amateur boxers have limits on the amount of bandage and tape allowed under the boxing glove, which diminishes the weight and momentum of their punches.⁴⁴ In professional boxing, there is generally no limit on hand wrapping, and heavier, more protective gloves are worn, increasing the weight and momentum of their punches. There are typically more injuries in competition at both levels because of the desire to maximize the forces of punches.

Facial Injuries

In a retrospective cohort study of competitive amateur boxers, the overall incidence of injuries was 24 for every 100 fights.⁴² Injuries including wounds and lacerations to the face and head represented 62% of injuries. In another study of professional boxers, the overall injury rate was 17 per 100 fights, with 51% of injuries involving wounds or lacerations to the face and head.⁴⁵ The other injuries evaluated were hand injury and eye and nose injuries. Injuries to the nose are very common often leading to epistaxis often impairing the boxers breathing during a match. The boxers with epistaxis should be evaluated for maxillofacial fractures, septal deviation, and possible septal hematoma.^{42,45} Repetitive trauma to the face can also lead to injuries to the eyes and surrounding structures. Soft-tissue injuries to the eyelids

can lead to the development of an expanding hematoma an effectively impair the sights of the boxer. Furthermore, direct impact to the eyes can lead to more serious conditions such as retinal detachment and orbital floor fractures. When suspecting a fracture, the evaluation of the athlete for impaired ocular movement, ptosis, and diplopia should be promptly completed.

Upper Extremity Injuries

Injuries to the hand and wrist are the most common musculoskeletal pathology encountered. Injuries to the hand represents 7% of all boxing injuries.⁴² In a cohort study of professional boxers, injury rate for hand and wrist injuries in competition was 347 injuries per 1000 hours, while the estimated injury rate in training was less than 1 injuries per 1000 hours.⁴⁶ The most common hand injuries observed were carpometacarpal instability and boxer's knuckle (neck of the fifth metacarpal fracture). Other fractures readily seen in the hand are metacarpal shaft fracture, and first metacarpal base fracture (Bennet fracture). Soft-tissue injuries involving the hand and wrist typically involve the scapholunate joint and interphalangeal dislocation. Most of these injuries necessitate a surgical intervention for stabilization. Soft-tissue injuries such as ligament strains are more readily seen than fracture.

Neurologic Injuries

Concussion represents from 10% to 17% of injuries seen in boxing and is the most common neurologic injury encountered.⁴² When a boxer is knocked down, there is a brief or prolonged loss of consciousness where the athlete is unable to stand or defend himself. In the majority of cases, the altered mental status lasts for a short period of time. There are cases of more severe trauma to the brain that can lead to death. Subdural hematoma are such cases and should be treated as emergencies.⁴² While a boxer can receive more than 100 punches to the head during a boxing match, the repetitive blows to the head through a boxer's career can lead to traumatic encephalopathy. It has been shown that about 17% of boxers can have a central nervous system lesion, which may have been caused by the repeated blow. Other studies have found mid to late stage chronic traumatic encephalopathy in 17% and cerebral or cerebellar atrophy or ventricular enlargement in 50% to 60% of the professional boxers.⁴⁷

AUTO RACING

Auto racing is a popular sport around the world that takes many different forms. In the United States, NASCAR is a popular sport where highly tuned stock cars are raced around an oval track at high speeds. In Europe and other parts of the world, auto racing takes the form of Formula 1, in which drivers of specialized racecars compete to go around a circuit track as quick as possible. Other forms of auto racing include rally car racing, stylistic driving competitions (e.g., drifting), and motorbike racing, each of which requires its own unique set of skills. In general, auto racing is considered a high-risk sport, as collisions can be morbid and fatal. While collisions are fairly common, safety standards have improved to decrease the injuries experienced from them.⁴⁸ Drivers also face a number of other injuries associated with the

sport.⁴⁹⁻⁵¹ Neck sprains and bruises are the most common injuries, presumably from the whiplash effect of the hard braking required in these sports and from the fact that the helmet increases the weight supported by the neck. Drivers may also experience wrist sprains, ankle sprains, tibia fractures, lumbar spine fractures, and abrasions around the body.⁴⁹⁻⁵¹

FIGURE SKATING

Figure skating is a unique and highly technical sport involving jumps, spins, footwork, dancing, and even acrobatics all on ice skates.⁵² There are four disciplines of the sport: singles skating, pair skating, ice dancing, and synchronized skating. The United States Figure Skating Association reports there are 680 skating clubs with more than 196,000 figure skating members in the nation.⁵³ The judging system tries to be as technical and objective as possible, and is composed of components such as speed of skating, transitions, performance/execution, choreography, amount of ice covered, and height of jumps.⁵² Due to the technical demand, figure skaters are susceptible to a number of injuries. Most of these are injuries are due to overuse.^{52,53}

Foot Injuries

Figure skates themselves consist of a stiff leather boot and a metal blade. The unique aspects include high heel of the boot—which places the foot into constant slight plantarflexion and the large toe pick on the front of the blade—which is used for some jump take-offs and jump-landings.⁵³ “Lace bite” is irritation of the tibialis anterior, extensor digitorum, or extensor hallucis longus tendons.⁵³ This occurs from excessive friction across the tendons, usually from improper placement of the tongue of the skate, and can be mitigated by ensuring proper tongue placement, or using boots with alternative lacing styles. “Pump bump” or Haglund's deformity is a protrusion of the lateral heel.⁵³ It is caused by an overly wide heel of the boot, allowing the skater's heel to move up and down, and can be mitigated by ensuring optimal boot heel fit. An accessory navicular is present in 4% to 21% of the normal population, excessive friction from a tight boot can make this a symptomatic process.⁵³ Treatment involves building up the arch of the skate, or having the area of the skate “punched out.” Stress fractures can occur most commonly in the first and second metatarsals.⁵⁴ They are caused by the force from jump take-offs, especially when engaging the toe pick of the skate. Treatment involves rest, screening for nutritional deficiencies, and in some cases implementing a “jump count” similar to a pitch count for stress injuries to baseball pitchers.⁵³

Ankle Injuries

Ankle injuries are the most common in all of figure skating.⁵⁵ They are usually sprains and in fact happen more commonly during “dry land” training as opposed to when on ice.⁵³ Malleolar bursitis can also occur from friction inside of the boot and is more common on the medial side.⁵³ Padding the affected malleolus or “punching out” the inside of the skate can mitigate the problem. Achilles tendinitis is common from overuse, thought to happen from the repetitive jumps, or compression from the rim of the boot.⁵³ Treatments include modifying the rim

of the boot, rest, ice, stretching, and eccentric strengthening of the posterior leg musculature.

Knee and Hip/Pelvis Injuries

Patellofemoral pain syndrome and patellar tendinitis can occur from jumping. Repetitive falls while learning new jumps and routines can lead to contusions throughout the lower extremity. Meniscus, ligamentous, and fractures of the lower extremity (outside of the foot and ankle) are actually quite rare.⁵³

Since many figure skaters will rotate in the same direction when performing jumps, they can develop an asymmetry in strength and flexibility.⁵³ Triple and quadruple jumps require a large amount of torque to achieve rotation speed commensurate with the jumps, and can lead to iliac crest apophysitis.⁵⁴ Since the iliac crest apophysis is one of the last to close (around age 16 in boys, and age 14 in girls), this can be an ongoing source of pain for teenagers.⁵⁶

Back and Upper Extremity Injuries

Arching trunk extension is a common motion in figure skating and can lead to a myriad of back conditions such as spondylosis, spondylolisthesis, lumbar strain, and facet joint pain.⁵⁷

Upper extremity injuries are rare, and occur more in pairs and dance skaters who have a higher risk of collision during practice and warm-ups compared with singles skaters. This can lead to lacerations, fractures, and even head injuries. These skaters also encounter shoulder and wrist problems resulting from the lifting and throwing elements in their discipline. Synchronized skating can be quite dangerous because of the number of skaters on the ice and the tight formations and maneuvers that they perform. A domino effect often occurs when one skater falls, creating a high risk for lacerations, finger amputations, fractures, and head injuries.⁵⁸

AMERICAN FOOTBALL

American football is a sport played between two teams of 11 players (in the United States) or 12 players (in Canada). The objective is to score points by either advancing the ball into the “end zone” of the field or kicking the ball through two raised goal posts. The team with the ball is considered on offense and the team without the ball is on defense. In the United States, the offense has four opportunities to advance the ball at least 10 yards to gain a new set of “downs.” In Canada, only three downs are allowed. The goal of the defense is to tackle the player with the ball and bring him to the ground to end a play (or “down”). Unlike basketball or soccer, players typically play only offensive or defensive positions. Each position is highly specialized, with a set of unique injuries that may occur.

American Football carries the highest injury rate of any team sport at all levels of play in the United States.⁵⁹ It is the most popular contact sport in North America, with more than 2 million athletes partaking in the sport in the United States alone.⁶⁰ Football players are susceptible to noncontact and repetitive stress injuries, and they are of course highly vulnerable to contact injuries. In some cases, as discussed below, football produces more contact-based mechanisms for injuries than in other sports

are commonly borne from noncontact mechanisms. A majority of injuries in football occur in the lower extremity, with a substantial portion of the remainder in the upper extremity. The most common types of injury are sprains and strains (40%), contusions (25%), fractures (10%), dislocations (15%), and concussions (5%).⁶¹

Head and Neck Injuries

Of most concern in this category are concussions. Concussions occur in an estimated 3 million youth athletes, 1.1 million high school athletes, and 100,000 college athletes each year; however, there is also an estimated 27:1 ratio of underreporting in college football, particularly among offensive linemen.⁶² New consensus guidelines exist to guide physicians and other health care providers.⁴⁰ Specific guidelines on when to remove athletes from play are critical, as adolescent and young athletes not removed from play have almost a 10-times risk of prolonged recovery as compared with those removed from play.⁶³ Though our recognition and treatment of concussions have come a long way, the barriers remain underreporting of injury, premature return to play (RTP), and receiving routine rather than individualized treatment.⁶²

Cervical spine injuries are uncommon, but a source of catastrophic injury in football. Due to recent rule changes modifying tackling and blocking techniques, the incidence has decreased—particularly the incidence of catastrophic injuries. Cervical spine injuries are composed of a spectrum of ligament and/or soft tissue damage, fractures, and neurologic impairment.⁶¹ Mechanism of injury is typically an axial load applied to a flexed or extended cervical spine.⁶¹ A “stinger” is neurapraxia of the brachial plexus or cervical spine nerve roots. It occurs in 50% to 60% of collegiate football players at some time in their careers.⁶⁴ The mechanism is either from neck hyperextension, lateral flexion, and axial load, or from a direct blow to the brachial plexus.⁶⁰

Upper Extremity

Upper extremity injuries are common in football, comprising about 30% of all injuries, with the shoulder affected most commonly.⁶⁵ Quarterbacks are most susceptible to shoulder injuries from throwing (overuse) or resulting from tackles while throwing (traumatic).⁶⁶ Shoulder injuries include acromioclavicular (AC)/sternoclavicular joint separations, acute rotator cuff injuries, repetitive strain injuries, subluxations/dislocation, and fractures.⁶⁶ Wrist sprains are another common injury, resulting from the heavy trauma to the distal upper extremity during tackling, blocking, and maneuvering other players on both offense and defense.⁶⁷ Fractures of the forearm and wrist are an additional common injury, usually resulting from impact with the ground while being tackled or with other players.⁶⁷

Lower Extremity

Lower extremity injuries are the most common type of injury in football. Starting in the thigh, muscle strains and contusions are extremely common. Contusions result from direct blows from the helmet, knee, or shoulder and can lead to bleeding within the musculature, swelling, pain, stiffness, and loss of muscle excursion limiting joint motion.⁵⁹ Muscle strains are common in muscles crossing two joints, as they experience higher stress

during eccentric contraction, with the quadriceps and hamstrings at particular risk.⁵⁹

Ligamentous injuries in the knee are the most common serious injury in football, and additionally the knee is the most common site of season-ending injury.^{68,69} Players injure their ACL with a reported incidence of 11 to 18 per 100,000 athlete exposures with increasing incidence at higher levels of play.⁷⁰⁻⁷² Valgus collapse of the knee is the common mechanism in both contact and noncontact injuries, with a forceful blow causing the former, and usually a sudden deceleration prior to change in direction causing the latter.⁷³ Though in many sports noncontact ACL injuries are more common, in football contact ACL injuries are in fact more common (55% to 60%).⁷³ Injury to the medial collateral ligament (MCL) is the most common knee injury at all levels of play, with estimated incidence of 24.2 per 100,000 athlete exposures in high school.⁷⁴ In a study examining collegiate athletes presenting to the National Football League (NFL) combine, 23% of offensive linemen had a history of MCL injury.⁷⁵ The mechanism (especially in linemen) is thought to be from “chop blocks” and other players “rolling up” on the outside of their legs.⁷⁴ Patellar subluxations and dislocations occur in 4.1 per 100,000 athlete exposures in high school.⁷⁶ The mechanism is commonly from result of knee flexion with the tibia in a valgus position. Though in general sporting most are from noncontact mechanisms, a majority of football instances (63%) are due to contact, similar to ACL injuries in football players.⁷⁶

Foot and ankle injuries have a reported incidence from 9% to 39%, with as many as 72% of all collegiate players presenting to the NFL combine having a history of foot and ankle injury, and 13% receiving prior surgical treatment.⁷⁷ Offensive and “skill position” players are particularly susceptible to foot and ankle injury from high levels of force and torque placed on the distal extremity during running, cutting, and tackling.⁷⁵ Lateral ankle sprain is the most common injury.⁷⁷ Turf toe is a hyperextension injury and plantar capsule-ligament sprain of the hallux metatarsophalangeal joint and is caused from increasing playing surface hardness and decreasing shoes stiffness.⁷⁷ Jones fracture is fracture of the fifth metatarsal at the metadiaphyseal junction, where there is a watershed area of decreased vascularity. The rising rate recently may be due to flexible, narrow cleats that don’t provide enough stiffness and lateral support for the fifth metatarsal during running and cutting.⁷⁸ Also, lateral overload from baseline cavovarus foot posturing with possible metatarsus adductus and/or skewfoot contributes as well.⁷⁸ Most of these are surgically fixed due to a delayed union rate of 25% to 66%, nonunion rate of 7% to 28%, and re-fracture rate of 33% in high-level athletes.⁷⁹ Lisfranc injuries are bony or ligamentous damage to the tarsometatarsal joints and occurs commonly from axial loading to a fixed plantar-flexed foot.⁷⁷ Noncontact twisting injuries are more common among NFL players, resulting in a purely ligamentous injury.⁷⁷ Comparison of weight-bearing radiographs of both feet is crucial to accurate and timely diagnosis.⁷⁷

GOLF

Golf is a sport in which an individual hits a small ball into a hole located hundreds of yards away with a club in as few strokes

as possible. A typical round of golf consists of 9 or 18 such holes of varying distances and topography. Each hole is graded on difficulty with a “par” number, which is the number of strokes it should take a skilled golfer to complete the course. Most holes fall between a 3 and 5 par. A sport requiring fine skill and concentration, golf is popular around the world and is played at all levels from friends engaging in a leisure activity to professional players competing in tournaments. Furthermore, as it is a light aerobic activity, people of nearly all ages can participate.⁸⁰

Injuries in golf are relatively common, as up to 40% of amateur players can experience a golf related injury.⁸¹⁻⁸³ While golf is not considered a high impact sport, the golf swing is a complex motion that involves the whole body. It requires rapid contraction and coordination of many muscle groups in order to accelerate the club to high velocities.⁸⁴ Factors that lead to the majority of golf injuries include poor mechanics, inadequate warm-up,⁸³ the seasonality of the sport, and occasional trauma. Furthermore, the swing can be performed 50 times or more⁸⁴ during a round, and this repetitive motion can lead to a number of injuries commonly seen in golfers.

The most commonly injured areas are the lower back (18.3%), elbow/forearm (17.2%), foot/ankle (12.9%), and the shoulder/upper arm (11.8%). Other rare but serious injuries related to golf cart trauma and environmental exposure can occur.^{82,85}

Lower Back Injuries

Low back pain is common among amateur and professional golfers. This can be primarily attributed to the biomechanics of the golf swing, which has evolved with changes in club technology to produce higher club head velocity at impact. The modern swing emphasizes greater rotation of the thorax through an arc of about 120 degrees. This places the vertebral column in a hyperextended position during the follow through phase of the swing and leads to large lateral flexion and anteroposterior forces on the lumbar spine.^{84,85} In addition, microtrauma to the hip leads to joint contractures and loss of rotation. Therefore, in order to adequately rotate the pelvis and torso for the swing, many golfers need to rotate the lumbar spine past its native rotational capacity. This leads to repetitive microtrauma, which predisposes golfers to paraspinal muscular strains, intervertebral disc herniation, and facet joint arthropathy. In concordance with these findings, most golfers report low back pain primarily during the follow-through phase and radiographs show more degenerative changes in lumbar facets of golfers compared with controls.^{81,82,84} Other contributors to low back pain can include carrying heavy bags and frequent bending on the course.^{83,86} Finally, many golfers with low back pain can rely on other muscle groups, such as the abdominal musculature, to achieve the same rotation, which can predispose these areas to overuse injuries as well.⁸⁴

Elbow/Forearm Injuries

Injuries to the elbow and forearm are the second most common in golfers after low back injuries.^{80,82} “Golfer’s elbow” is the term applied to medial epicondylitis, which typically occurs when the golf club strikes the ground forcefully during the downswing. This traumatic injury typically occurs in players with poor swing

mechanics.^{85,87} Lateral epicondylitis, on the other hand, is typically an overuse injury due to repetitive forceful forearm extension along with twisting and is actually five times more common than “golfer’s elbow.”^{85,87} Other potential elbow injuries include ulnar neuritis and medial epicondyle avulsion fractures.⁸⁷

Shoulder Injuries

The shoulder is next most common area of upper extremity injury. The lead shoulder (left in a right-handed golfer) is the most commonly affected side.^{80,88} The rotator cuff is active with elevating the club above the head for the address phase of the swing, and players can experience symptoms of subacromial impingement with this activity. Pain from the AC joint can also manifest during this phase of the swing, especially during maximal cross body adduction when forces along the AC joint are greatest.⁸⁸ In order to generate more power during the downswing phase of the swing, many golfers will rotate their shoulder joint as much as possible, which can lead to microtrauma to the joint capsule and labrum, leading to symptoms of glenohumeral instability. Posterior instability is more common than anterior, and many golfers will have a positive load and shift test on exam. SLAP tears and biceps pathology are also possible, but less common than other true overhead sports.^{80,84,85,88}

Wrist/Hand Injuries

Wrist and hand injuries can be traumatic or due to chronic overuse. Instability of the extensor carpi ulnaris tendon can result with forceful striking of the club into the ground during the downswing. Patients will complain of a snapping sensation on the ulnar side of their wrist. Hook of the hamate fractures can occur with traumatic strike of the club and typically occur in the lead hand. Patients will experience tenderness at the hook with gripping the club and may have ulnar paresthesia symptoms.^{80,81,85,89} Chronic overuse injuries include triangular fibrocartilage complex (TFCC) tears, triquetrum-hamate impingement, pisotriquetral arthritis, and ulnar styloid impingement. These injuries occur due to the excessive ulnar deviation and wrist extension during the golf swing. They can be difficult to differentiate without advanced imaging, as they all present with ulnar-sided wrist pain and tenderness and pain with gripping the golf club.⁸⁹

GYMNASTICS

Gymnastics is an Olympic sport that has more than 5 million participants per year in the United States.⁹⁰ While more than 75% of participants are female, the men and women athletes participate in different events.⁹¹ The women participate in four categories: vault, uneven parallel bars, floor, and beam performance. The men participate in six events: vault, parallel bars, high bar, ring, floor, and pommel horse. As in other sports discussed in this chapter, gymnasts are subjected to a higher frequency of injury in competition than in practice.⁹¹ An observational study of collegiate athletes found that female gymnasts are twice more likely to sustain an injury during competition than in training.⁹⁰ Furthermore, the same study found that the majority of injuries were lower extremity injuries sustained during the vault or floor

exercises.⁹⁰ While events such as the vault are proponents of lower extremity injuries because of one acute landing following an acrobatic jumping maneuver, the floor exercise possess an increased risk with the sheer number of landings during a normal routine. Other events such as the parallel bar (uneven and even) and the rings place the upper extremities at risk with placing the shoulder in vulnerable position with swings, handstands, and isometric contractions; moreover, the constant release and regrip of the bars adds to the complexity of the routine and increases exposure to injury. Of note, the majority of athletes participating in gymnastics are skeletally immature children and thus they are subject to injuries to their growth plates. The gymnast wrist is the most frequent injury encountered in the growing child. It entails an injury to the distal radial physis secondary to the repetitive stress from the axial loads to the hand and wrist. It has been reported that gymnast wrist has an incidence of 1.9 to 2.7 injuries per 100 participants per season.⁹²

Lower Extremity Injuries

The most common injury encountered in gymnastics is during landing and therefore involves the lower extremity. This type of injury represents around 69% of all gymnastic injuries.⁹¹ The technical complexity of landing, coupled with the high energy of an acute landing, place the lower extremity at high risk of injury. While ankle sprains are the most frequent injury encountered, injuries to the knee come second including those involving the anterior cruciate or collateral ligaments.⁹⁰ The repetitive bounding and landing place the foot at risk of overuse injuries such as calcaneal apophysitis, sesamoiditis, planter fasciitis, and stress fracture. Furthermore, the routines are executed barefoot, and that places the foot at risk for nerve entrapment syndrome.⁹⁰ The pathology of the knee is due to overuse and can lead to chronic injuries of the knee including patellofemoral syndrome, patellar tendonitis, and Osgood-Schlatter disease in addition to acute injuries of the cruciate and collateral ligaments.

Upper Extremity Injuries

Musculoskeletal injuries to the upper extremities are readily seen both in male and female gymnasts. In males they are the most common injury and more often involve the glenohumeral joint.⁹³ In females, the upper extremity is the second most common site of injury and most often involves the wrist and elbow.⁹³ The glenohumeral joint pathology includes dislocation and labral tearing from instability. These athletes are often hyperlax and vulnerable to shoulder instability. Other pathology in the shoulder are rotator cuff tendinosis from overuse and rotator cuff impingement.^{90,93} The injuries to the wrist and elbow include fracture of the long bones, collateral ligament injuries, and overuse injuries to the apophysis and ligaments.

Back and Pelvis Injuries

Lastly, lumbar spine pathology is often encountered in gymnasts with the repetitive jumps and landing during the majority of the events. There is a prevalence of 25% to 85% of lower back pain ranging from strains, herniated discs, to spondylosis and stress fractures due to flexion and hyperextension of the back. The treating physician should be aware of stress fracture that

may mimic an array of injuries and should investigate menses history and eating habits.

ICE HOCKEY

Ice hockey is a sport practiced on a sheet of ice, which is surrounded by a perimeter of boards and glass. It is a rectangular surface with rounder corners and its dimensions may vary depending the geographical location. In North America, the surface is smaller with dimensions of 200 ft × 85 ft (61 m × 26 m)⁹⁴ compared with the wider international dimensions of 200 ft × 98 ft (61 m × 30 m).⁹⁵ The game is played with each team having six players on the ice at a given time, which comprise of three forwards, two defensemen, and one goaltender. An ice hockey game is played with three 20-minute periods where players compete to shoot a rubber puck with a stick into the opponent's net in order to score points. A team typically has 20 players on the bench with forwards and defensemen alternating on ice play every minute and each team with goaltenders playing the entire game.

The game of hockey is played worldwide by youth and adults alike and is sanctioned as a Winter Olympic sport. The rules of ice hockey differ between males and females in that body checking is not permitted in women's hockey.⁹⁶ The age where body checking is allowed in men's hockey varies depending on regional regulation. A recent youth hockey study found a threefold increased risk of all game-related injuries in youth hockey where body checking was allowed.⁹⁷ The players wear full body protective equipment along with skates with metal blades, which enable them to glide on the ice surface. The protective equipment is similar across women's and men's hockey with the exception of facial protection. Females and under-18-year-old males must wear full-facial protection, but men over 18 can wear partial-facial protection or no facial protection in certain professional leagues.⁹⁸ The injuries encountered in ice hockey are predicated around the following: it is a contact sport, players can move as fast as 45 km per hour and shoot the puck as fast as 190 km per hour, and players use a metal composite stick while wearing skates with sharp metal blades.

Head Injuries

While players wear head protective gear, concussions are readily encountered in this fast-paced contact sport where players are often hit while in vulnerable positions. Concussions are the most common injury encountered, with head and neck injuries representing between 20% and 30% of all injuries.⁹⁹ Facial laceration, contusions, and maxillofacial injuries are seen when full-facial protective equipment is not mandated or when fighting is allowed.¹⁰⁰ The incidence of concussion ranges from 0.2/1000 to 6.5/1000 game-hours and 0.72/1000 to 1.81/1000 athlete-exposures, and is estimated at 0.1/1000 practice-hours.¹⁰¹ The injury prevention efforts surrounding concussion has been the subject of many changes by the regulating bodies of ice hockey. To this effect, injury prevention efforts have been successful in focusing on improved and proper use of equipment,^{102,103} changes of on-ice rules,^{104,105} and educational prevention programs.^{103,105} The injuries preventions education program has drastically

affected the incidence of spinal injuries when body checking an opponent from behind is banned.¹⁰⁶ Furthermore, the risk of concussion was significantly lower if games were played on rinks with flexible boards and glass.⁹⁸

Upper Extremity Injuries

The upper extremities injuries represent around 8% to 20% of all injuries, with the majority involving glenohumeral dislocation and AC dislocations.¹⁰⁷ Since ice hockey is performed with a hockey stick and is sometimes used to stop or prevent an opponent from progressing, it also inflicts a significant amount of trauma to the upper extremities. Fractures around the hand and wrist are inflicted either by contact with a stick or with a puck, even with adequate protection because of the high velocity generated. Lastly, players can sustain rib fractures and contusion from contact with the boards after receiving a body check from an opponent.

Lower Extremity Injuries

Injuries to the lower extremities represent between 20% and 30% of injuries in ice hockey.¹⁰⁷ There is a constellation of hip injuries that are specifically encountered in ice hockey.¹⁰⁸ These include extraarticular injuries such as hip adductor, flexor muscles strain, hip pointers, and also core muscle injury. Intraarticular injuries such as acetabular labral tears and femoral impingements are also prevalent in this athletic population.¹⁰⁸ Furthermore, fractures of the proximal femur or the long bone and contusion of large muscles groups are also encountered in light of the collision nature of the sport. The injuries involving the knee include the meniscus and most frequently the MCL. Lastly, high ankle sprain and fracture around the foot and ankles are often encountered from receiving a high velocity shot. The blade of the hockey skate can often inflict significant trauma to the blood vessels or traumatic laceration to the soft tissue such as the Achilles tendon. Furthermore, there are documented tracheal injuries from players receiving the puck in the neck area.

TEAM HANDBALL

Team handball is played on an indoor court with a layout similar to a soccer field. There are two goals, one on either end. Two teams compete, with seven on each team, composed of one goalie and six court players.¹⁰⁹ The object of the game is to score by throwing the ball into the goal of the opposing team. Players can dribble the ball or pass similar to basketball, and are only allowed to take three steps with the ball in hand otherwise.

Handball is one of the Olympic sports with the highest risk of injury.^{110,111} The game is characterized by high playing tempo, rapid changes in movement, jumps with hard landings, frequent contact/collision between players, and repetitive knee and shoulder joint stress.¹¹² The overall incidence of injury is about 104 to 108 injuries per 1000 player-hours, or about 1.5 injuries per match.¹¹³ In surveillance at the world championship in 2015, 27% of players suffered at least one injury.¹¹³ A majority of injuries occur during match play. Also a majority are in the lower extremity (58%), as compared with the upper extremity (17%), head/face (13%), and torso (12%).¹¹³ Most

injuries were classified as contusions (39%)—especially to the face, thigh, knee, and lower back—followed by ligament sprains (24%), and muscle strains (13%), especially to the thigh and groin areas.¹¹³ Most injuries were a result of contact between players (61%), with less occurring due to noncontact trauma (16%), or overuse (12%).¹¹³

Shoulder Injuries

Shoulder injuries comprise a substantial portion of all injuries in handball. Shoulder pain is most common in senior handball, with a prevalence estimated at 19% to 36% at season start, and a weekly prevalence of 28% during the season.¹¹⁴ The prevalence in the elite adolescent population is lower (about 14%, or 1.4 per 1000 player-hours); however, 80% of these involve the dominant (throwing) arm.¹¹⁵ A significant risk factor is a large increase in playing load of 60%, conferring a hazard ratio of 1.91 in this population versus those with a decrease or small-to-moderate increase in load.¹¹⁵ This is also exacerbated by reduced shoulder external rotation strength and scapular dyskinesis.¹¹⁵

Facial/Dental Injuries

Facial trauma in handball is caused mainly by blows to the facial area with hands or elbows during collisions between players due to the high-speed and physical style of play.¹¹⁶ In addition, some injuries are caused by being hit by the ball at close range.¹¹⁷ Most nondental facial injuries are lacerations (19%), followed by nose injuries (18%), and eye and periorbital injuries (16%).¹¹⁸ Dental injuries account for 22% of facial injuries, consisting mostly of socket bleeding, tooth fracture, tooth avulsion, and tooth luxation.¹¹⁸ Dental injuries in handball carry a 76% complication rate, most commonly resulting in tooth color change, pulp necrosis or infection, and potentially culminating in tooth loss.¹¹⁸ Mouthguards have been demonstrated to significantly reduce dental injuries and decrease jaw fracture risk.¹¹⁸ Though the American Dental Association has classified handball as a sport requiring use of mouthguards, use is still rare.¹¹⁹

LACROSSE

Field lacrosse is played on an outdoor field setup similar to a hockey rink. There are two goals, one on either end, and space behind the goal is utilized as well. Two teams compete, with 10 on each team, composed of one goalie, three defensive players, three midfielders, and three offensive players. The offensive and defensive players from each team must remain in their respective areas, while the midfielders are allowed to freely traverse both sides of the field.¹²⁰ The object of the game is to score by launching the ball from the lacrosse stick into the goal. Players can advance the ball by passing it in the air or on the ground to their teammates, or can “cradle” the ball in their stick and run with it. The game is played for a set amount of time, after which the winner is the team with more points.

Men's and women's lacrosse are the two fastest growing contact sports in the United States.¹²¹ The National Collegiate Athletic Association (NCAA) has noted a constant increase in varsity men's programs in the past decade from 211 schools and 7103

student-athletes in 2003 to 2004 to 350 schools and 13,165 student-athletes in 2014 to 2015.¹²² The injury rate is 5.29 per 1000 athlete-exposures, just over half of which are time-loss injuries.¹²¹ The lower extremity is most commonly injured (58%)—especially involving the knee.¹²¹ The upper extremity is also commonly involved (21%)—with the arm/elbow causing the highest proportion of non-time-loss injuries.¹²¹ The trunk (11%) and head/neck (9%) are less frequently involved.¹²¹ The most common injury types are sprains (25%), strains (25%), and contusions (17%).¹²¹ Contact-related mechanisms accounted for 50% of injuries, with noncontact (36%) and overuse (11%) contributing to a lesser extent.¹²¹ The top three specific injuries include ankle sprains, knee internal derangement injuries, and concussions.^{40,121,123}

Ankle Injuries

Ankle injuries are typically caused from noncontact cutting, dodging, or torsional activities.¹²⁴ Fortunately, these injuries are typically not very severe, and confer no significant time loss or complications in most cases. Prevention strategies are being developed; presently they include evaluation for preexisting ankle instability during the preparticipation examination, complete rehabilitation and recovery following injury, and taping/bracing to protect the athlete from further injury in addition to adequate rehabilitation.¹²⁴

Knee Injuries

Knee internal derangements, specifically ACL injuries, are a common injury in lacrosse.¹²⁴ These injuries are typically caused from noncontact, cutting, and pivoting, causing indirect injury to the knee.¹²⁴ Since these injuries are often severe and result in significant time loss from sport, injury prevention strategies and programs are being developed. These incorporate balance training, plyometrics, strengthening, and feedback to alter biomechanical and neuromuscular contributing factors.^{124,125}

Concussions

Lacrosse carries the fourth and third highest rates for concussions at the collegiate level in men's and women's sports, respectively.^{126,127} Concussions are most commonly caused by player-to-player contact in men's games and from incidental contact with the lacrosse stick in women's games.¹²⁴ Management is comprehensive, and includes preseason planning, education, and initial evaluation. This is then compared with postconcussion assessment and disposition. From this, RTP decisions can be made more appropriately and accurately, to achieve an optimal long-term neurological outcome. Baseline assessments are becoming a crucial factor in the above protocol.¹²⁸ A study utilizing a nonhelmeted mastoid-patch accelerometer to collect head impact data demonstrated fortunately that a vast majority of head impacts in lacrosse do not result in concussion.¹²⁶ However, the authors note that the hypothesized short- and long-term detrimental effects of “subconcussion” on brain function and structure, coupled with their proposed role in increasing susceptibility to neurodegenerative disorders, suggest that quantification of “subconcussion” in sport may be more important in the future.¹²⁶

ROWING

Rowing is a competitive sport that has been practiced for centuries and was one of the inaugural Olympic sports. It is practiced by amateurs and professionals, and is a popular collegiate sport as well. Rowing is an aerobic and anaerobic exercise with great cardiovascular benefits. Competitive rowing is divided into sculling, which is the use of two oars, and sweep rowing, in which athletes use one oar on opposite sides to propel the boat. Training is typically done on an ergometer, which is a stationary rowing machine.¹²⁹

Understanding the rowing stroke is paramount to proper understanding of the injuries that can occur in this sport. The first phase is the drive in which the athlete is in the “catch position” where the arms are extended, and the trunk and the knees are flexed. As the oar is lowered into the water, force is applied through the body from the feet through extension of the knee and then extension of the trunk to drive the oar forward. This phase finishes with flexion of the arms to propel the boat forward. Next is the recovery phase, in which the oar is extracted from the water and rotated (“feathered”) so it is parallel to the ground. In reverse order from the drive, arms extend, the trunk flexes, and the knees flex. Injuries typically occur from overuse and overtraining on the ergometer.¹³⁰

Lumbar Spine Injuries

Low back pain is the most common musculoskeletal complaint among rowers with up to 53% of athletes with this complaint.^{131,132} The rowing stroke requires lumbar spine hyperflexion and twisting forces with transmission of up to 4.6 times the rower’s mass in compressive load. The repetitive nature of rowing combined with excessive training can lead to fatigue, impaired muscle contractility, impaired proprioception, and altered kinematics, all of which increase the risk of muscular or ligamentous strain in the lumbar spine.¹³⁰ Furthermore, the cyclical compressive loading can lead to disc herniation, facet capsule strain, and even end plate compression fractures. Studies have also shown a higher risk of lumbar spine spondylolysis among rowers compared with the general population, likely due to the rotational forces seen.^{129,131}

Chest Injuries

Rib stress injuries are a relatively common injury with an incidence of up to 9% and are responsible for most of the missed time for athletes.^{129,132} Ribs 5 to 9 are most commonly involved, and the mechanism is thought to be an imbalance between microtrauma from repetitive forceful muscle contraction and ability of the bone to remodel and heal in response to this stress. The exact mechanism is unknown, but several theories have been postulated, including opposing forces of the serratus anterior and external oblique muscles and ribcage compression during force transmission to the oar.¹³⁰ Costochondritis and intercostal muscular strains can also occur, typically due to the oar striking the thorax at the end of the stroke.¹³⁰

Upper Extremity Injuries

Shoulder injuries occur from altered glenohumeral mechanics. The humeral head can be anteriorly shifted, the posterior capsule

may be tight, the latissimus dorsi may be tight as well, and the rotator cuff muscles can become weak. The combination of these changes can lead to symptoms of impingement and instability. Sweep rowers experience unilateral shoulder pain on the lead side.¹³⁰ Wrist and forearm injuries typically arise from improper mechanics, excessive wrist motion during feathering the oar, tight grip, or use of the elbow rather than the shoulder to initiate drive phase of the stroke. Injuries include exertional compartment syndrome, De Quervain’s tenosynovitis, intersection syndrome, and lateral epicondylitis. “Sculler’s thumb” is a rowing specific injury in which improper feathering using the thumb leads to hypertrophy of the extensor pollicis brevis and abductor pollicis longus causing compression of the underlying radial extensor tendons.^{129,130}

Knee Injuries

Knee pain is common complaint among rowers.¹³² However, as rowing is not a weight-bearing activity and does not involve pivoting, acute ligamentous and meniscal injuries are typically not seen. Patellofemoral syndrome accounts for the majority of knee pain in this population. During the rowing stroke, the knee goes through the full ROM. The rower starts with the knee in full deep flexion when the patellofemoral joint experiences high compressive loads. As the knee is extended, patellar maltracking can occur due to valgus position of the knee during the drive phase. The combination of these two can lead to chronic patellofemoral pain.¹³⁰ Some rowers can experience lateral knee pain from iliotibial (IT) band syndrome as the IT band moves across the lateral femoral condyle in a repetitive fashion.^{129,130}

RUGBY

Rugby is a form of football where two teams of 13 or 15 players attempt to score a touchdown by carrying or kicking the ball downfield. Players are able to pass the ball either sideways or backwards to each other. Positions include forwards, whose role it is to contest the ball, and backs, whose role it is to carry the ball and score touchdowns. A certain number of passes is allowed before the other team is granted the ball and a chance to score. The teams can fight over possession during a “scrum,” which occurs after each play stoppage. After a tackle, teammates protect the ball by forming a platform, which the opposite team attempts to break. Unlike American football, players only wear shoulder and chest pads and scrum caps.¹³³

Due to the large amount of collisions and relatively little protective gear, musculoskeletal injuries are relatively common in rugby. Most injuries are acute and occur in the lower extremity. Forwards also experience more injuries than backs. In the lower extremity, the knee is the most common location, and MCL sprains/tears are the most common. ACL tears represent a serious injury, causing players to miss a significant amount of time. Backs are at higher risk for hamstring injuries, as they need to sprint and stop quickly during the game.¹³⁴

Shoulder injuries can occur from direct contact with other players, especially during a scrum. AC joint injuries are the most common, but some players can experience traumatic shoulder dislocations, which are more serious.¹³⁵⁻¹³⁷

During the scrum, players are at risk for spinal injuries. With their heads flexed, an axial load can lead to a cervical spine injury. The hooker is the player at the tip of the pyramid, a position that puts him at higher risk for cervical spine injuries. These players typically have their arms wrapped around other adjacent players in the scrum, so they are less able to shift their body and shield themselves from damaging their C spines. Lumbar spine injuries are also possible, but occur more in the training season as an overuse injury rather than an acute traumatic one.^{133,134}

Head injuries are also seen in rugby players. Given the physical and sometimes violent nature of the sport, players put themselves at higher risk for closed head injuries. While headgear can protect against serious head injuries, they have not been shown to prevent the most common mild injuries.¹³⁸

RUNNING

Running is a very common activity that has been shown to provide significant cardiovascular health benefits. Practiced at a wide variety of levels, running can be a leisurely activity or highly competitive sport. Running encompasses many different forms of activity, including endurance running in marathons, sprinting, and hurdling, each of which requires a different set of skills and has different associated injuries.¹³⁹ Lower extremity injuries, as expected, account for the grand majority of complaints in runners and are typically chronic overuse type injuries. Acute injuries can occur in runners, but are much less common than in other sports.¹⁴⁰ Running injuries are complex, and significant sex differences exist. Risk factors for injuries in runners include little running experience and running distances greater than 32 km/week.¹⁴¹

The knee is the most common site of injury, accounting for up to 50% of running injuries. Patellofemoral pain is the most common knee injury followed by IT band syndrome, and meniscal injuries. Females tend to have higher rates of patellofemoral knee pain, while males tend to have more meniscal injuries.^{140,141}

Overuse injuries can occur below the knee as well and include Achilles tendinopathy, medial tibial stress syndrome, exertional compartment syndrome, and plantar fasciitis. Thin (BMI <21 kg/m²) runners with poor nutrition can have metatarsal stress fractures as well. Blisters can also develop in areas of high friction.^{139,140}

More serious injuries related to the environment can occur as well. Marathon runners can experience heatstroke and dehydration. Running in extreme cold temperatures can lead to frostbite or hypothermia.¹³⁹

SKIING AND SNOWBOARDING

Skiing

Skiing is a popular sport and leisure activity across the world with an estimated 200 million skiers. Skiers place long thin boards on their feet to allow them to slide down a mountainside. Shifting the person's weight on the skis allows them to turn. They use poles to help keep balance and push off at the start. In the most common competition, downhill skiers attempt to navigate a course on the mountainside formed by poles in the least amount of time. Other forms exist as well, such as cross-country skiing

and high jump. Advances in ski technology as well as course maintenance have allowed skiers to go faster and have increased the popularity of the sport, but also increased the resultant incidence of injury.¹⁴²

Knee Injuries

Knee injuries account for a large proportion of all skiing injuries due to the high rates of speeds and failure of the boot bindings to release, leading to a rotational force on the knee.¹⁴³ Isolated MCL sprains are the most common injury, but ACL tears are the most significant. Downhill skiing has one of the highest rates of ACL tears, matching that of American football. Skiers can also have associated knee injuries along with ACL tears, including MCL and meniscal tears. Female skiers, as in other sports with high rates of ACL injuries, have significantly higher rates of ACL tears than males.¹⁴³ Prevention strategies have been implemented, such as improved equipment and educating skiers on the common mechanisms of ACL tears and how to position themselves appropriately in those situations. Other acute injuries can occur as well, such as isolated meniscal tears. PCL injuries and tibial plateau fractures are generally rare in skiing.^{142,143}

Shoulder and Thumb Injuries

The most common upper extremity injury in skiing is "skier's thumb," which is a sprain of the thumb UCL at the metacarpophalangeal joint.¹⁴² Shoulder injuries are also relatively common in skiing and occur primarily due to axial load from falling on to the extremity, but can also occur through direct trauma to the shoulder or from wrist straps on the poles, which can force the shoulder into abduction and external rotation. The commonly seen shoulder injuries include rotator cuff strains, glenohumeral dislocations, AC separations, and clavicular fractures.¹⁴⁴

Other Skiing Injuries

The "boot top" fracture is a tibia/fibula fracture that is less common now with higher boots that have release bindings. Other sprains, fractures, and lacerations have been reported, but are less common in skiing.^{142,144}

Snowboarding

Compared with skiing, snowboarding is a younger sport with a smaller but still sizable number of participants. Competition in snowboarding is similar to that in skiing, but snowboarders also participate in style events such as the halfpipe, in which aerobatic maneuvers are judged on execution and style. These events require high jumps and put snowboarders at elevated risk for higher energy mechanism injuries.¹⁴⁴

Upper Extremity Injuries

In contrast to skiing, upper extremity injuries are more common in snowboarding. This is due to the fact that snowboarders tend to fall more often and without poles they then use their outstretched hands to break their fall.¹⁴⁵ Furthermore, the boots worn are fixed to the board and are nonreleasable, so there is less lower extremity movement than in skiing. Wrist injuries include distal radius fractures, carpal fractures, sprains, and contusions. Shoulder injuries include glenohumeral dislocations,

clavicle fractures, AC separations, rotator cuff tears, and proximal humerus fractures.^{142,144,145}

Lower Extremity Injuries

While lower extremity injuries are less common than upper extremity, there has been an increasing incidence of them due to advances in equipment and courses that allow snowboarders to go faster, thus leading to higher energy injuries. These injuries tend to occur more often in the lead extremity.¹⁴⁵ Ankle injuries are the most common, and fractures about the ankle are also the most common lower extremity fracture. An injury specific to snowboarding is the “snowboarder’s ankle” or a fracture of the lateral process of the talus. These are relatively rare injuries but are important, as they are often missed on plain X-rays and often misdiagnosed as ankle sprains. Knee injuries are much less common in snowboarding compared with skiing and often occur when only one foot is attached to the board.^{142,145}

Injuries in Both Skiing and Snowboarding

Experienced skiers and snowboarders may attempt to traverse steeper slopes and uncharted courses, which can put them at higher risk for head and spinal injuries, which are less common than other sports but are a cause of significant morbidity. Recent high-profile cases have shed light on the dangers of skiing at high speeds. Skiers and snowboarders can collide with various obstacles along the course such as trees, rocks, pylons, and lifts. Helmets are now commonplace on ski slopes. They may also fall onto their backs or necks leading to spinal injuries.¹⁴²

SOCCER

The sport of soccer, also known as football for the rest of the world, is the most widely practiced sport worldwide. In Europe alone, there are more than 20 million licensed soccer players.¹⁴⁶ Both female and male players participate in this sport, which is sanctioned by the Olympic program. Soccer is played either indoors, or more traditionally, outdoors. The outdoors field dimensions range from 64 to 75 m long by 100 to 110 m wide, and at the adult level, it is played with a circular ball around 22 cm in diameter. The two opposing teams have 12 players on the field, including a goaltender on each end of the field. The game is played with two 45 minutes halves of running time where the opposing team competes to score the most goals by using any part of the body except the upper extremities. Soccer is classified as a collision sport; however, body checking is not permitted. The players are permitted to challenge the opponent while jumping for headers or by slide tackles targeted at the ball. During a professional soccer game, it is estimated that a player will run an average of 10 kilometers. While it is considered an aerobic sport, the frequent stop and go and sprints bring the average intensity of close to the anaerobic threshold with heart rate reaching 80% to 90% of its maximum.¹⁴⁷

The incidence of soccer injuries has been widely studied in professional and nonprofessional athletes in virtually all continents of the world. In a cohort study of eleven professional clubs in five European countries, the mean injury incidence was 9.4 injuries per 1000 hours, with 30.5 injuries per 1000 match hours

and 5.8 injuries per 1000 training hours.¹⁴⁶ The risk of injury during matches can be eight times greater than in practice.^{146,148} Men and women have the same incidence of injuries with an increased in injury rate with age.^{148,149} Another variable that influences the incidence of injury is the type of surface of the soccer field. While earlier studies had shown an increased in injuries on artificial turf field, recent studies comparing newer artificial turf with natural turf have not shown any differences. The majority of injuries are traumatic in nature, where 41% are caused by contact and 59% are caused by noncontact mechanism.¹⁵⁰ Furthermore, 25% of injuries that occur during matches are directly caused by foul play.¹⁴⁶

Lower Extremity Injuries

In soccer, the overall majority of injuries affect the lower extremity. In a cohort study of a professional European soccer team, Walden et al. found that 85% of injuries were located in the lower extremity, where the most common type of injury was thigh strain.¹⁴⁶ Hamstring strain was more prevalent than quadriceps strain, followed adductor muscles strain. About 50% of the ligament sprains were located in the ankle and 40% in the knee. The most common knee ligament injured was the MCL.¹⁴⁶ Other injuries encountered in the knee ACL tear, meniscus tear, and patellofemoral syndrome. ACL tears account for 14% of injuries in soccer.¹⁵¹ There are differences in the rate of ACL injuries based on sex. Recent studies have estimated that women are six times more likely to sustain an ACL tear in a noncontract mechanism than men.¹⁵¹ Since soccer is mainly played with the feet, there is a wide array of different mechanisms that create foot and ankle trauma. Low and high ankle sprains can occur when the foot is planted and subjected to axial load with external rotation, or while the player is landing from a header. While ankle strains are the most prevalent pathology of the foot and ankle, fractures and stress fractures are often encountered. Furthermore, overused tendinopathy of the Achilles tendon is also observed frequently.

Upper Extremity Injuries

Upper extremity injuries are more frequent in youths than in adults. These injuries can include fractures of the arm, wrist, hand, and clavicle. Specific to youth are the apophyseal disorders that can include Osgood-Schlatter disease; Sever disease; Sinding-Larsen-Johansson disease; and apophysitis of the anterior superior iliac spine, anterior inferior iliac spine, and iliac crest of the pelvis. Acute avulsion fractures to apophyses (anterior inferior iliac spine, tibial tuberosity, and lesser trochanter) can occur, along with growth plate fractures.

Concussions

Concussions can represent up to 9% of injuries and are caused by collisions between players while jumping for headers, or when the head strikes the ground or objects like the soccer ball or the goal post. The remaining head and neck injuries are typical of sports where accidental collision is frequently encountered. Maxillofacial injuries such as fracture, laceration, contusions, and abrasion are sustained due to the lack of facial protective equipment. The U.S. Soccer Federation released concussion guidelines

in 2016 to reduce risk especially in the youth population. These guidelines prohibit heading the ball for children age 10 and younger, and limit the amount of heading permissible in practice for children between the ages of 11 and 13.¹⁵²

Young Athlete Injuries

Youth soccer also involves specific type of injuries. Fracture involving the growth plates, along with avulsion fractures of the apophysis, is encountered. Furthermore, overuse injuries can affect the youth soccer player with pathologies such as jumper's knee, Osgood-Schlatter, or patellofemoral chondrosis or low back pain from spondylolisthesis. As in other sports, educational injury prevention programs have been efficient in decreasing the incidence of injury. These programs often target neuromuscular training and functional dynamic strengthening of core muscle strength.¹⁵¹ The Federation Internationale de Football Association (FIFA) 11+ is a program that was effective at reducing the incidence of ACLs per year in youth female soccer with neuromuscular training and strengthening exercises.¹⁵³

SWIMMING

Swimming is a popular sport enjoyed by all generations and all levels, from recreation to fitness to competition. Though the buoyant effect of water seems to protect swimmers from impact-type injuries, injuries still occur from overuse, repetitive strain, and microtrauma mechanisms.¹⁵⁴ Competitive swimmers at a club level may swim about 10 km/week, whereas elite international competitors may swim upward of 100 km/week.¹⁵⁵ In a NCAA survey, elite swimmer injury rates were about 4 per 1000 hours of training.¹⁵⁶ The difference between swimming and most other sports is that the propulsion or advancement through the water is generated from the upper extremity more than the lower extremity.¹⁵⁵ Not surprisingly, the most common injury is to the upper extremity—specifically to the shoulder as the keystone of the upper extremity conveying the propulsive force.

Shoulder Injuries

Shoulder injuries are the most common swimming injury, with a reported prevalence of 40% to 91%.¹⁵⁷ Elite swimmers may swim more than 9 miles per day, which translates to more than 2500 shoulder revolutions.¹⁵⁵ “Swimmer's shoulder” was initially described as anterior shoulder pain during and after workouts, and its cause postulated to be from impingement of the rotator cuff tendons under the coracoacromial arch (outlet impingement).¹⁵⁸ However, it became evident that swimmers have multiple factors contributing to their shoulder pain, and “swimmer's shoulder” is now a catchall term for any shoulder pain in a swimmer. The mechanisms driving this shoulder pain include poor stroke biomechanics; overuse and fatigue of muscles of the shoulder, scapula, and upper back; and glenohumeral laxity with subsequent shoulder instability.¹⁵⁷

Knee Injuries

Knee injuries are the second-most common injury in swimming. “Swimmer's knee” is an equally ambiguous term as “swimmer's shoulder,” coined for any knee pain in a swimmer. Its reported

prevalence ranges from 34% to 86%.¹⁵⁷ It also goes by “breast-stroker's knee,” indicating its high prevalence in this group; in fact breaststrokers are five times more likely to have “swimmer's knee” as compared with all other swimmers.¹⁵⁹ Breast-stroke imparts a high valgus load to the knee secondary to an adducted hip position. The valgus strain placed on the MCL of the knee is further exacerbated during the “whip kick.” Affected swimmers present with medial-sided knee pain, either along the length or insertions of the MCL or overlying the pes tendons and/or bursa. Of note, the “flutter kick” of the freestyle swimmers involves much less coronal plane strain, resulting in a 10-times lower risk of developing “swimmer's knee” as compared with breaststrokers.¹⁵⁹ However, the repetitive “flutter kick” can lead to increased patellofemoral contact stresses from the forceful and repetitive quadriceps contractions, resulting in anterior knee pain. The incidence of knee pain in freestyle swimmers is still low despite this.

Back Injuries

All swimming strokes maintain hyperextension of the lower back to achieve a streamlined position; this is exaggerated in the “undulating” style of breaststroke and butterfly, making these swimmers at high risk for spondylolysis and possible spondylolisthesis.¹⁶⁰ It is estimated that the prevalence of low back pain in breaststroke swimmers is 22% to 47% and in butterfly swimmers is 33% to 50%.¹⁶¹

TENNIS

Tennis is a sport in which a racket is used to hit a small ball across a net with the goal of keeping the ball within the court lines during back-and-forth rallies until either player hits the ball out of the court lines or into the net. It is the most popular racquet sport and is a very popular activity worldwide. It is practiced as a leisure activity as well as a professional sport. A tennis match can be contested either between two individuals (“singles”) or between two groups of two athletes (“doubles”).

Biomechanically, the racquet strokes in tennis involve muscles in the whole body, which allows the generation of high racquet head velocities. The serve is an important part of the game and can be thought of as a kinetic chain that starts from the feet and transmits force through the whole body into the racquet. Large loads are generated especially at the elbow and shoulder joints, which helps explain the pattern of injuries commonly seen.¹⁶²

Injuries in tennis can be to either the lower or upper extremity. Lower extremity injuries tend to be more acute in nature, while upper extremity injuries tend to be chronic overuse injuries.¹⁶³

Lower Extremity Injuries

Ankle injuries tend to be the most common in the lower extremity due to the rapid change of direction required during rallies. Different playing surfaces also likely place different strains on the ankle, but these differences have not been explored in the literature. Ankle ligament sprains resulting from inversion mechanisms are the most common acute injury.^{162,163} Players may also develop peroneal subluxation, as the peroneal muscles are the

primary restraint to ankle inversion and thus the tendon sheath can become weak with repetitive inversion forces. Other ankle injuries include anterior impingement, subtalar instability, and osteochondral lesions. Hip injuries can also occur as the serve and stroke require forceful hip rotation to generate sufficient power. Muscle and ligament strains are the most common, and other less common injuries include labrum tears and posterior impingement. Repetitive and forceful acceleration can lead to Achilles tendon, gastrocnemius, and hamstring tears. Common knee injuries include patellar tendinitis and patellofemoral syndrome. Other injuries include plantar fasciitis, stress fractures, and tennis toe (repetitive impact of second and third toes on the shoe).^{162,163}

Upper Extremity Injuries

As tennis is an overhead racquet sport, upper extremity injuries are quite common. Lateral epicondylitis or “tennis elbow” occurs from the torque generated at the racquet, which leads to microtrauma at the extensor tendon insertion. Vibrations from the ball hitting the strings are also transmitted to the forearm and contribute to this microtrauma. Advances in technology have made racquets lighter, allowing higher velocities during the tennis swing and as a result higher torque and vibrations transmitted to the forearm.¹⁶² Other common elbow injuries include flexor-pronator tendinitis, medial epicondylitis, ulnar neuritis, and posterior impingement. Shoulder injuries are also relatively common. The injuries with the highest incidence include glenohumeral internal rotation deficit (GIRD), rotator cuff tears, and SLAP tears.¹⁶³ Wrist injuries include extensor carpi ulnaris tendinitis and subluxation. Players often complain of ulnar-sided wrist pain.¹⁶⁴

Low Back Injuries

Low back pain is a frequent complaint among tennis players. The lumbar spine is subjected to considerable axial load and torque during the tennis serve and swing. These repetitive motions can fatigue the ligaments and muscles around the lumbar spine, leading to sprains and tears. Rotational forces and hyperextension of the lumbar spine can also lead to intervertebral disk degeneration and degenerative disk disease.¹⁶²

VOLLEYBALL

Volleyball is a sport that typically takes two forms: court and beach. In court volleyball, two teams of six players play on a court separated by a high center net. One player serves the ball from one end of the court over the net to the other team to start the rally. The goal is to keep the ball from hitting the ground and using only the player's arms/hands to propel the ball and create shots. Players on the receiving team may “bump” the ball over the net or “set” the ball up for a teammate with a high bounce so that he/she may “spike” the ball downward with increased force. Other ways to hit the ball are to “dig” the ball up just before it hits the floor and to “block” the ball at the net before it crosses over. The team that allows ball to hit the ground on their side of the net, hits the ball out of the court, or hits the ball more than three times loses the point.¹⁶⁵

Beach volleyball has the same basic rules as the court variety, but is played on a sand court outdoors. The teams also consist of only two players. There are other rule differences between the two types of volleyball, but they share the same shots and the same point structure. Both beach and court volleyball are practiced at a variety of levels, including at high-level professional and Olympic events.¹⁶⁵

The characteristics of volleyball play require high jumping and using the whole upper extremity in an arc motion to accelerate the ball to high velocities. Therefore injuries can be seen to both the upper and lower extremity in players.^{165,166} While injuries in volleyball are less common than those in contact sports such as football, the incidence of injuries has been increasing, likely due to an increase in the number of amateur players.¹⁶⁷ Injury patterns differ between beach and court volleyball. Beach volleyball players experience more overuse injuries, whereas court volleyball players have a higher incidence of acute injuries.¹⁶⁵

Ankle Injuries

Ankle sprains are the most common acute injury in volleyball players, with studies estimating an incidence of up to 40% of all injuries. Sprains typically occur when a player jumps and lands with his/her foot on the foot of another player, leading to an inversion injury. These injuries occur more frequently in court volleyball due to the higher number of players on the court and the hard surface they play on.¹⁶⁶⁻¹⁶⁹ Prevention strategies that have been shown to decrease the incidence and recurrence of ankle sprains include rule changes, improved landing techniques, neuromuscular training, and the use of ankle braces and taping.^{168,169}

The repetitive jumping and landing in volleyball can also lead to Achilles tendinitis in the acute setting and tendinosis, typically in older players as an overuse injury. Players will complain of tenderness and crepitus about the Achilles tendon.¹⁶⁵⁻¹⁶⁷

Knee Injuries

The most common knee injury seen is patellar tendinopathy, which results from the explosive jumping. Some studies have estimated the incidence of acute patellar tendinitis to be about 40%. Overuse tendinopathy or “jumper's knee” can occur due to the deep flexion angle of the knee that players experience while jumping and landing.^{166,170,171} Beach volleyball players experience less patellar tendon pathology than court volleyball players, likely due to the differences in playing surfaces.¹⁶⁵

ACL tears are less common in volleyball compared with other sports such as basketball, but are more serious than other injuries. The mechanism for ACL tears is similar to other sports. Players may land awkwardly or pivot while landing, which can lead to a large valgus moment and eccentric loading of the knee. In general, players with ACL tears are unable to continue playing and have to miss a significant amount of time to undergo ACL reconstruction. Many athletes are unable to return to their previous level of play.^{165,168}

Finger/Wrist Injuries

The high velocity of the ball can lead to many acute and chronic injuries of the fingers and wrist, especially in blockers who attempt to stop the ball with their hands as it crosses the net. The axial

and hyperextension load of the ball can lead to finger fractures and IP joint dislocations. Displaced fractures or irreducible dislocations require a referral to an orthopedic surgeon for evaluation. Soft tissue injuries can also occur including volar plate injuries, jersey fingers (distal interphalangeal [DIP] joint hyperextension leading to FDP injury), and ligamentous injury to the PIP joint. Thumb radial collateral and UCL damage can also occur when the thumb is forced into abduction during blocking or falling to the ground. Falling on the outstretched hand while attempting to perform a “dig” can lead to hook of the hamate fractures.^{165,167,172}

While returning the serve, the ball repetitively hits the volar aspect of the wrist. This microtrauma can lead to De Quervain's tenosynovitis, inflammation of the first dorsal compartment of the wrist.¹⁶⁵

Shoulder Injuries

The glenohumeral joint involves a large humeral head articulating with a relatively small glenoid. This gives it a large ROM, but also makes it an inherently unstable joint. The ligaments, rotator cuff muscles, and capsule of the shoulder work together to provide stability to the joint in normal daily activities. During the volleyball serve and spike, the shoulder is brought into the extremes of this ROM, and the resultant microtrauma with each swing leads to damage of these elements and resultant shoulder pain. The arm swing involves three phases: cocking, acceleration, and deceleration. The initial cocking phase places the arm in the extremes of external rotation and extension, which can lever the humeral head anteriorly.^{167,171,173} The rotator cuff normally prevents subluxation or dislocation of the humeral head, but with repeated swings, overuse of the rotator cuff can lead to strains and tears of the cuff muscles and tendon. This position of the humeral head can also lead to tearing of the labrum or inferior glenohumeral ligament, or internal impingement. Shoulder dislocations can occur as well, typically when the player attempts a “dig” and lands on an outstretched hand.^{165,172}

Suprascapular nerve palsy is also fairly common, with many volleyball players complaining of periscapular pain and weakness. Various theories exist as to the etiology. The nerve can become entrapped in the spinoglenoid notch by a ganglion, which can develop after a labral tear. Another theory postulates that the extreme ranges of motion cause traction on the nerve, and with the repeated swings, a nerve palsy develops.^{165,167,170}

GIRD may also develop from repetitive arm swings. The posterior capsule becomes tightened in order to prevent anterior subluxation of the humeral head, restricting the ability of the glenohumeral joint to internally rotate.¹⁶⁵

Back Pain

Low back pain can occur in volleyball players when the spine is repetitively flexed and rotated during the swing. The force is primarily transmitted through the pars interarticularis and stress reactions can occur. Players typically complain of back pain localized to the facet joints. Although not common, bilateral pars defects can lead to spondylolisthesis with possible neurologic deficits, which require a surgical consultation. Disc pathology

is also less common, but can occur due to the repeated spine flexion.^{165,166,172}

Concussion

Concussions are less common in volleyball compared with collision sport athletes, but have been reported in the pediatric age group. These injuries typically occur from collision of the player's head with the pole or ground. Preventative measures have not been studied in the literature.¹⁶⁵

WRESTLING

Wrestling is the world's oldest sport, boasting hand-to-hand combat with the ultimate goal of pinning the opponent's back to the mat. Points are awarded for moves that gain control over the opponent, or bring him closer to being pinned on his back. Two main styles exist internationally: Freestyle and Greco-Roman. The main difference between these is that in Greco-Roman, attacks to the legs are prohibited, encouraging upper body maneuvers, such as headlocks, throws, and suplexes. In the United States, Folkstyle is the main style contested in the youth, high school, and collegiate levels. In Folkstyle there are no additional points awarded for throws as compared with traditional takedowns, additionally there are no points for quickly exposing the opponents back, only for holding the opponent on their back in a controlled fashion. These differences de-incentivize wrestlers to perform more dangerous moves allowed and rewarded in the international styles, which thus lessens the overall injury incidence in Folkstyle wrestling.¹⁷⁴ Nevertheless, injury rates in high school are reportedly 2.33 to 2.50 per 1000 athlete exposures.¹⁷⁵ Injury rates increase in college to 7.25 per 1000 athlete exposures.¹⁷⁴ In the 2008 Beijing Olympics, 9.3% of wrestlers participating in the event sustained an injury.¹⁷⁶ Since both wrestlers are on their feet and more mobile in the neutral position, the majority (39% to 42%) of injuries occur in this position.¹⁷⁴

Musculoskeletal Injuries

The shoulder is the most commonly injured upper extremity joint, followed by the elbow and wrist. Shoulder injuries make up the most (18% to 24%) single joint injury in high school wrestlers.¹⁷⁴ Subluxation and dislocations are common and often precipitated by extreme forward flexion, abduction, and external rotation occurring either actively or passively.¹⁷⁴ These events can be associated with rotator cuff strain or labral tear, predisposing the wrestler to subsequent recurrent injury with less force. One study found that wrestlers with generalized ligamentous laxity experienced half the number of injury as those without, postulating that a “stiff” shoulder is injured more commonly than a lax shoulder.¹⁷⁷

The elbow is commonly injured when a defensive wrestler is being returned to the mat from a standing position, falling onto an outstretched arm while catching himself. Dislocations in this setting are common, with most (80%) being posterior and having significant disruptions of supporting ligaments.¹⁷⁴

Wrist and hand sprains are common with either the hand or wrist becoming caught on an opponent or the mat during various attack and evasive maneuvers. Finger dislocations are especially

common.¹⁷⁴ If dislocated in the coronal plane, then one or both collaterals are damaged, and if in the sagittal plane, then the volar plate is likely injured.¹⁷⁸ The PIP joint is the most common finger joint to be dislocated.

The knee is the most commonly injured lower extremity joint, followed by the ankle. Sprains to collateral ligaments, especially the MCL are common. These are precipitated by various defensive maneuvers during takedown from neutral position, as well as self-imposed valgus loads to the knee when a defensive wrestler is slowly standing against an offensive wrestler's force. Meniscal injuries are also common, as significant force is often encountered across an unstable knee in a flexed position from driving into an opponent, or standing up against significant resistance. Prepatellar bursitis can also develop from repetitive trauma of a wrestler striking their own knee on the mat during takedowns, and maneuvering defensive positions. A small percentage of these will go on to infectious bursitis and can cause time loss from sport. Ankle ligament sprains also occur frequently.

Cervical spine injury is usually noncatastrophic, and composed mainly of cervical strains and sprains, though fractures and other trauma can occur.¹⁷⁴ These are precipitated by defensive techniques and counter moves. Catastrophic injuries are rare, but tend to occur more so at the high school level.¹⁷⁹ Catastrophic injuries occur almost exclusively when wrestlers are in the neutral position, during takedown maneuvers.

Head injuries can result from collision with an opponent's head, knee, elbow, or the mat. Concussion is the primary form of head injuries seen in wrestlers. It should be recognized and treated as with other sports, with swift recognition and removal from further participation crucial and the responsibility of the covering health care provider.

Soft Tissue Injuries

Lacerations are common in wrestling from contact with opponents' head, elbow, knee, and/or fingers/nails. The eyebrow is a common site for laceration, and though different methods exist to quickly stop the bleeding to allow the match to continue, the best is to suture the area when time permits.¹⁷⁴

Auricular hematoma (or "cauliflower ear") is very common in wrestlers. The process involves blunt trauma to the ear, causing an accumulation of blood in the subperichondrial space, resulting in hematoma formation.¹⁸⁰ This interrupts the blood supply to the cartilage, and if the hematoma is not removed, the cartilage will ultimately die. Subsequently, new abnormal cartilage forms with fibrosis, resulting in the typical "cauliflower" deformity.

Epistaxis is the most common cause of bleeding in wrestling. The dry winter air during the season makes nasal mucosa more friable and prone to bleeding. The most common site is Kiesselbach's plexus.¹⁷⁴ As with lacerations above, it is important for the bleeding to be controlled quickly to minimize match delay, which in some cases can lead to disqualification. Application of cotton rolls can keep pressure on the area during the match, and vasoconstrictive sprays are helpful after the match.

Skin Infections

Skin infections are the most common condition for wrestlers seeking medical attention, and the single condition accounting

for the most time loss from practice and competition.¹⁸¹ The primary risk factors are localized breakdown in the natural skin barrier from repetitive trauma and exposure to skin pathogens. Herpes simplex virus has a predilection for the head, neck, and face, which suggests that skin-to-skin contact between wrestlers is a major vehicle for transmission. In examining skin conditions preventing wrestlers from participating in competition, 45% were due to herpetic whitlow, 25% from bacterial infections, and 22% from fungal infections.¹⁷⁴ It is important to wrestlers frequently self-inspect as early detection can minimize time loss, spread to other wrestlers, and duration of infection.¹⁷⁴

Weight-Loss-Related Issues

Wrestlers commonly reduce their weight to gain a strength advantage wrestling at a lower weight class. Wrestlers will not lose the weight and maintain throughout the season, however, but will instead cycle their weight 5 to 10 lbs. in a single week before competition.¹⁸² Depending on the season, this could translate to gaining and losing that amount of weight 10 to 30 times per season. The common techniques utilized are sweating and fluid or caloric restriction. Sweating is induced by exercise in hot environments with excessive clothing or impermeable suits. Less commonly used methods include consumption of diuretics or laxatives, and self-induced vomiting. In the fall of 1997, three collegiate wrestlers died within a 5-week period from rapid intentional weight loss.¹⁸³ Since this unfortunate event, wrestling governing bodies have implemented regulations of weight loss to mitigate the growing problem. Wrestlers are prohibited from cutting weight below a level of 5% body fat in college or 7% body fat in high school. In addition, certification in the beginning of the season involves hydration testing and body fat calculations, if over the aforementioned minimums, wrestlers are allowed to reduce their weight by 1.5% per week to reach their final weight. In addition, weigh-ins are held usually within 1 hour of the start of competition, and not the night before competition.

For a complete list of references, go to [ExpertConsult.com](https://www.expertconsult.com).

SELECTED READINGS

Citation:

McCroly P, Meeuwisse W, Dvorak J, et al. Consensus statement on concussion in sport—the 5th international conference on concussion in sport held in Berlin, October 2016. *Br J Sports Med*. 2017.

Level of Evidence:

V

Summary:

This article is the most recent consensus statement on the diagnosis and management of sports-related concussions. They cleverly institute 11 Rs to cover all aspects of the paper: Recognize, Remove, Re-evaluate, Rest, Rehabilitation, Refer, Recover, Return to sport, Reconsider, Residual effects and sequelae, Reduction.

Citation:

Kinsella SD, Thomas SJ, Huffman GR, et al. The thrower's shoulder. *Orthop Clin North Am*. 2014;45(3):387–401.

Level of Evidence:

I

Summary:

This article represents a concise review of the thrower's shoulder.

Many aspects of shoulder pain in throwing athletes can be understood better from knowing about the phases of throwing, the different muscle groups involved, and where poor mechanics may start to cause pathology.

Citation:

Dines JS, Bedi A, Williams PN, et al. Tennis injuries: epidemiology, pathophysiology, and treatment. *J Am Acad Orthop Surg*. 2015;23(3):181–189.

Level of Evidence:

I

Summary:

This article reviews common injuries sustained by tennis players.

The authors begin by discussing the tennis serve and stroke including the biomechanics and kinetic chain that are used as the basis to explain the mechanisms of injury. They also discuss the advances in equipment that have changed the patterns of injury seen. Each area of the body is then discussed and the epidemiology and mechanisms of injury are reviewed.

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